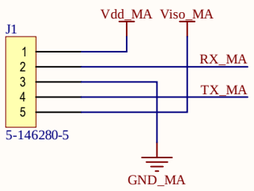
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<SCH DWG NO> DWG. NO. | REV | 1 SHT.

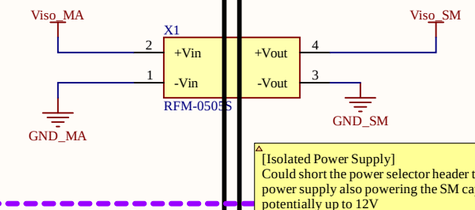
REVISION	DESCRIPTION	DATE	APPROVED

Information Side

Input Connector

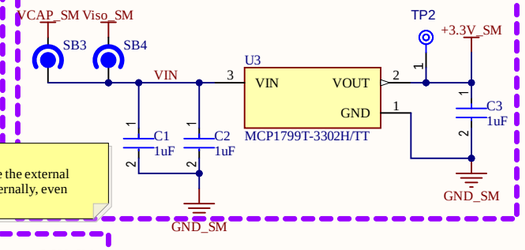


Isolated Supply

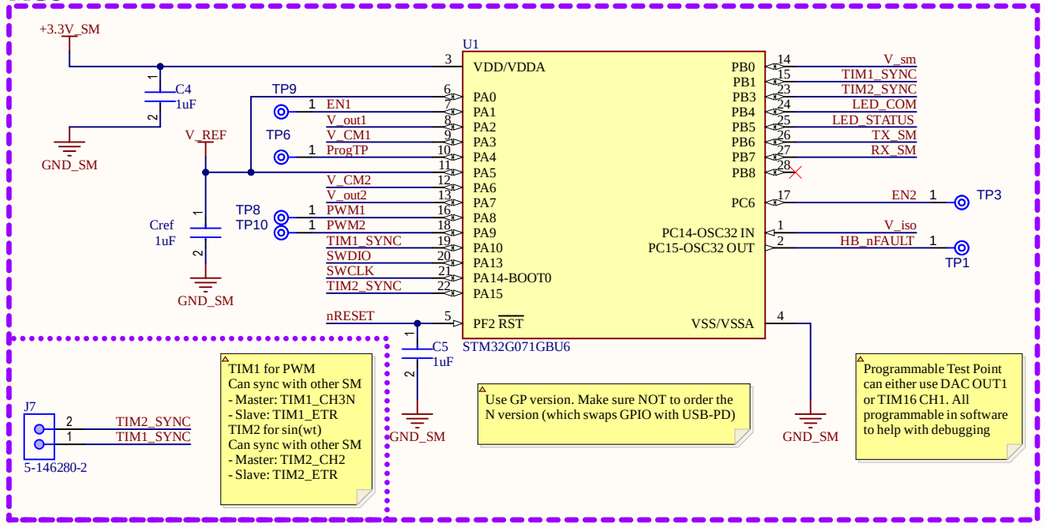


[Isolated Power Supply]
Could short the power selector header to have the external power supply also powering the SM caps externally, even potentially up to 12V

5V/3.3V Regulator



MCU



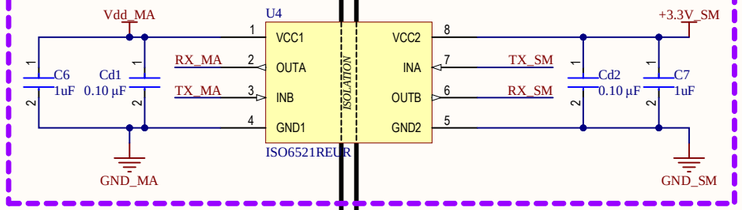
[TIM1 for PWM
Can sync with other SM
- Master: TIM1_CH3N
- Slave: TIM1_ETR
TIM2 for sin(wt)
Can sync with other SM
- Master: TIM2_CH2
- Slave: TIM2_ETR

Use GP version. Make sure NOT to order the N version (which swaps GPIO with USB-PD)

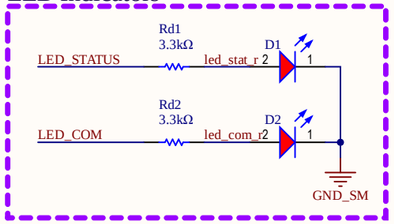
Programmable Test Point can either use DAC OUT1 or TIM16 CH1. All programmable in software to help with debugging

J7
TIM2_SYNC
TIM1_SYNC
5-146280-2

Digital Isolator



LED Indicators

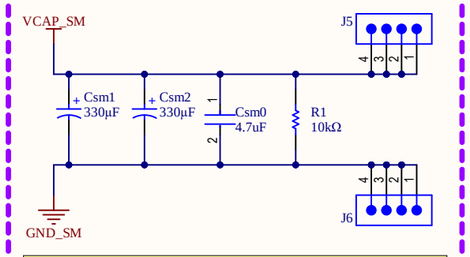


Isolated (Master - MS) Side

Floating (Sub-Module - SM) Side

Power Side

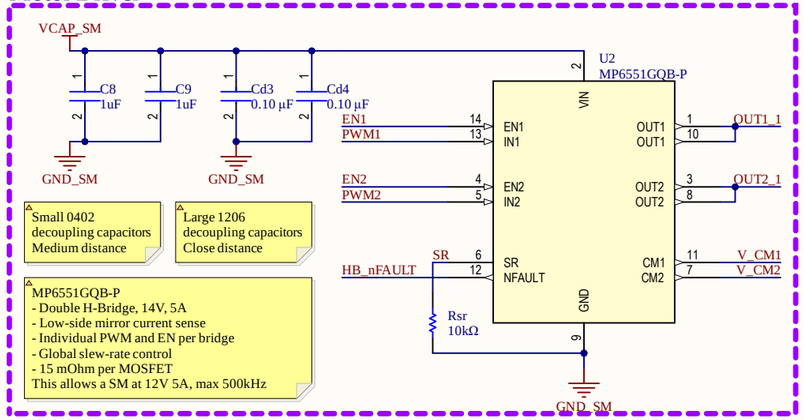
SM Capacitor



[Sub-Module Capacitance]
Energy requirement: 0.05J/W
Each SM deals with 10W [in practice 7mF nominal]
Too much capacitance on the main SM board,
Thus split it across two boards:
- Main PCB will have 2x 330uF
- Extra capacitance (5-10mF) added externally to reach full potential

[Self-Discharge]
T = 6*RC = 1min
R = 600s/(6*1mF) = 10kOhm
[in practice 10k / 2]
Pr = 12V^2/5k = 29mW

Motor Driver

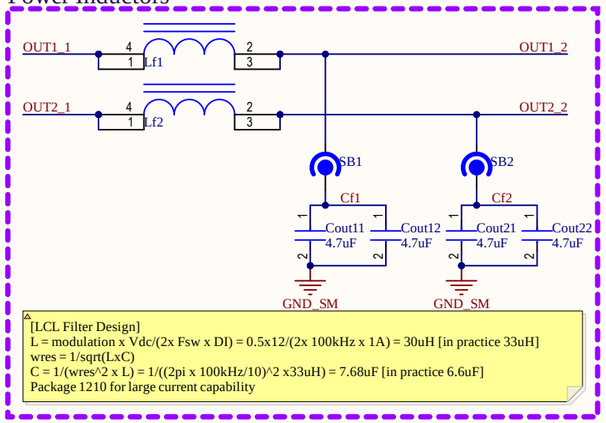


Small 0402 decoupling capacitors Medium distance

Large 1206 decoupling capacitors Close distance

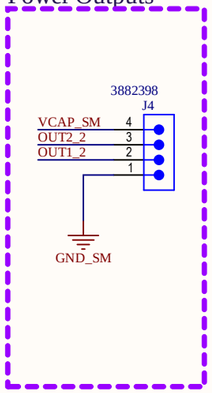
MP6551GQB-P
- Double H-Bridge, 14V, 5A
- Low-side mirror current sense
- Individual PWM and EN per bridge
- Global slew-rate control
- 15 mOhm per MOSFET
This allows a SM at 12V 5A, max 500kHz

Power Inductors

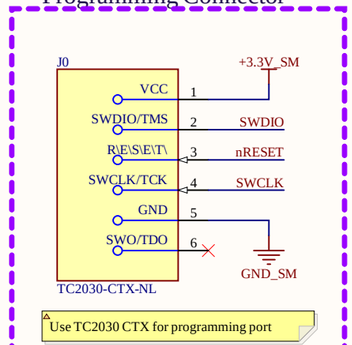


[LCL Filter Design]
L = modulation x Vdc/(2x Fsw x DI) = 0.5x12/(2x 100kHz x 1A) = 30uH [in practice 33uH]
wres = 1/sqrt(LxC)
C = 1/(wres^2 x L) = 1/((2pi x 100kHz/10)^2 x33uH) = 7.68uF [in practice 6.6uF]
Package 1210 for large current capability

Power Outputs



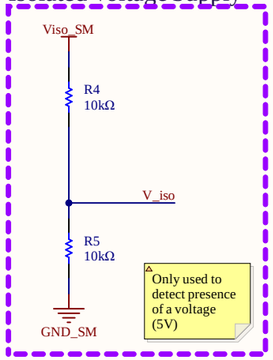
Programming Connector



Use TC2030 CTX for programming port

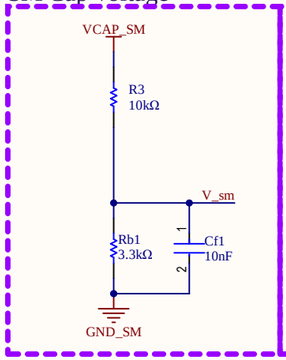
Measurement Side

Isolated Voltage Supply

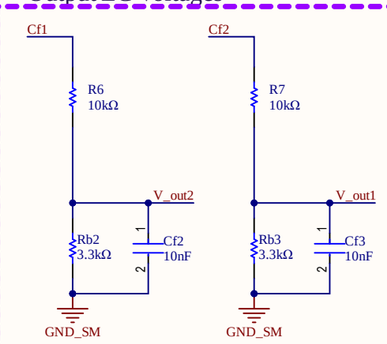


Only used to detect presence of a voltage (5V)

SM Cap Voltage

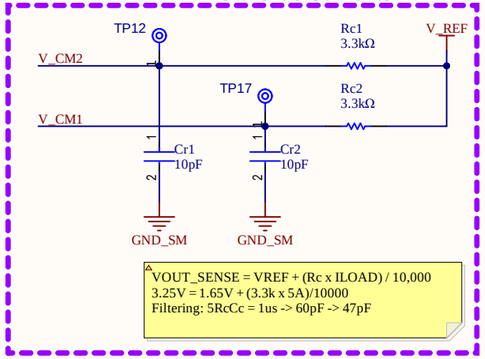


Output LC Voltages



[Voltage Measurement]
Vmax = 14V, Vadc = 3V
Rlow = 3.3k, Rhigh = 10k
wc = 1/RC = 100kHz / 16
C = 10nF

Low-Side Currents



VOUT_SENSE = VREF + (Rc x ILOAD) / 10,000
3.25V = 1.65V + (3.3k x 5A)/10000
Filtering: 5RcCc = 1us -> 60pF -> 47pF

Mechanical



APPROVALS		DATE
ENG:	MMC Merlin	--/--/--
DSN:	MMC Merlin	--/--/--
CHK:	MMC Merlin	--/--/--
REFERENCE DOCUMENTS		
BOM:	<BOM DOC NO>	
ASSY DWG:	<ASSY DWG NO>	
FAB DWG:	<FAB DWG NO>	
PCB DWG:	<PCB DWG NO>	

PROJECT: MicroMMC_S
Altium
2175 SALK AVENUE
SUITE 200
CARLSBAD, CA 92008
USA
PROJECT REVISION: Version control disabled
DOCUMENT REVISION: Version control disabled
DESIGN ITEM:
TITLE: Micro-MMC Sub-Module V3

SIZE	CAGE CODE	DWG NO.	REV
B	?????	<SCH DWG NO>	

SCALE: | FILE NAME: Top_Schematic.SchDoc | SHEET 1 OF 1

<SCH DWG NO> DWG. NO. | REV | 1 SHT.